

	MODULE <i>DieHard</i>	
EXTENDS <i>Integers</i> VARIABLES <i>small, big</i> $TypeOK \triangleq \wedge small \in 0 \dots 3$ $\wedge big \in 0 \dots 5$ $Init \triangleq \wedge big = 0$ $\wedge small = 0$ $FillSmall \triangleq \wedge small' = 3$ $\wedge big' = big$ $FillBig \triangleq \wedge big' = 5$ $\wedge small' = small$ $EmptySmall \triangleq \wedge small' = 0$ $\wedge big' = big$ $EmptyBig \triangleq \wedge big' = 0$ $\wedge small' = small$ $SmallToBig \triangleq$ IF $big + small \leq 5$ THEN $\wedge big' = big + small$ $\wedge small' = 0$ ELSE $\wedge big' = 5$ $\wedge small' = small - (5 - big)$ $BigToSmall \triangleq$ IF $big + small \leq 3$ THEN $\wedge big' = 0$ $\wedge small' = big + small$ ELSE $\wedge big' = small - (3 - big)$ $\wedge small' = 3$ $Next \triangleq \vee FillSmall$ $\vee FillBig$ $\vee EmptySmall$ $\vee EmptyBig$ $\vee SmallToBig$ $\vee BigToSmall$		